

Your grade: 100%

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Next item →

1. What information do we need in order to design a Kalman filter? (Select all that apply.)

1 / 1 point

- ☐ We need to have manufacturer part numbers for the sensors that will be used with the Kalman filter.
- ☒ We need to describe what we know about the uncertain process noise w_k and sensor noise v_k .

✔ Correct
Yes, this is true.

- ☒ We need a discrete-time state-space model of the dynamics of the system.

✔ Correct
Yes, this is true.

- ☒ We need to describe what we know about the uncertain initial state x_0 of the system.

✔ Correct
Yes, this is true.

2. What is the full output of a Kalman filter at any given timestep?

1 / 1 point

- ☒ The estimated value of the state vector at this timestep, plus confidence bounds on this estimate.
- ☐ The predicted value of the state vector at this timestep.
- ☐ The predicted value of the state vector at this timestep, plus confidence bounds on this prediction.
- ☐ The estimated value of the state vector at this timestep.

✔ Correct
Yes. This is correct.

3. Which of the following are properties of the Kalman-filter solution? (Select all that apply.)

1 / 1 point

- ☒ The prediction step uses the known dynamics of the state and known characteristics of the process noise to predict the state's values.

✔ Correct
Yes, this is true.

- ☒ The correction step combines the prediction and the measurement and their uncertainties to make a state estimate valid for this timestep.

✔ Correct
Yes, this is true.

- ☐ The Kalman-filter's state estimate will converge exactly to the true state if the filter is allowed to run long enough.

- ☒ The Kalman filter produces confidence bounds on its estimate that allow us to interpret the output of the filter appropriately.

✔ Correct
Yes, this is true.